

## Mock Test — Particulate Nature of Matter

A mixed paper covering the whole chapter. Total: 20 marks. Section A is interactive.

### Objective (1 mark each)

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**Q1 • MCQ****[1]**

The smallest units that make up a substance are its —

- (a) constituent particles (b) mixtures (c) bubbles (d) spaces

**Q2 • MCQ****[1]**

Which has a definite volume but no definite shape?

- (a) solid (b) liquid (c) gas (d) none

**Q3 • MCQ****[1]**

Interparticle attraction is strongest in —

- (a) gas (b) liquid (c) solid (d) equal

**Q4 • MCQ****[1]**

Diffusion is fastest in —

- (a) solids (b) gases (c) liquids (d) all same

**Q5 • MCQ****[1]**

Which is easily compressible?

- (a) gas (b) liquid (c) solid (d) none

**Q6 • TRUE / FALSE****[1]**

Dissolving sugar in water is a chemical change.

- (a) True (b) False

**Q7 · FILL IN THE BLANK**

**[1]**

The gaps between particles are called \_\_\_\_\_ spaces.

\_\_\_\_\_

**Q8 · FILL IN THE BLANK**

**[1]**

The temperature at which a solid melts is its \_\_\_\_\_ point.

\_\_\_\_\_

### **Short answer (2 marks each)**

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**Q9**

**[2]**

State the shape and volume properties of solids, liquids and gases.

\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_

**Q10**

**[2]**

Why are gases compressible but solids and liquids are not?

\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_

**Q11**

**[2]**

Define diffusion and state where it is fastest.

\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_

### Long / application (3 marks each)

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Q12

[3]

Explain, using particles and interparticle forces, the conversion of a solid to a liquid to a gas on heating.

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Q13

[3]

Explain (i) why sugar dissolving in water hardly changes the water level and (ii) why the smell of incense spreads across a room.

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[Check yourself](#) → Answer key